

Parameter Estimation For Model Based UR3 Robot Arm control

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Chapter 1: Introduction & Problem Statement

1.1 The Benchmark Platform: A 6-DOF Legacy Manipulator

- Structurally based on a 6-Degree-of-Freedom serial manipulator with an industrial joint configuration.
- Consists of 6 sequential revolute joints utilizing a modified proximal Denavit-Hartenberg frame convention.
- Link mass distributions, center of mass coordinates, and structural inertias are entirely uncalibrated and unknown.

1.2 The Core Problem: The Pitfalls of Model-Free Control

- Conventional standalone Proportional-Derivative control loops suffer from severe position tracking degradation under dynamic loads.
- Unmodeled dynamic variations and gravitational forces pull the link masses downward away from desired trajectories.
- Reactive controllers require structural tracking errors to generate opposing torques, making high-precision paths impossible.

1.3 The Rationale: The Necessity of Parameter Estimation

- **Unavailability of Direct Measurement:** True link masses, centers of mass, and inertia tensors cannot be directly measured or verified without completely dismantling the physical robot structure.
- **Inaccuracy of CAD Baselines:** Theoretical factory data ignores variations from age, structural wear, retrofitted components, and internal manufacturing tolerances.
- **Mathematical Non-Invasiveness:** System identification via parameter estimation represents the only feasible method to extract these values purely from non-invasive joint position and torque input-output data.

1.4 The Objective: Full Base Parameter Estimation and Virtual Control Validation

- Establishing a comprehensive system identification pipeline to estimate the complete dynamic parameters of the manipulator.
- Deriving a full, linear-in-the-parameters symbolic regressor model to isolate the 48 minimum base dynamic parameters.
- Validating the mathematical accuracy of the entire estimation process by deploying a standalone gravity torque compensation controller within a high-fidelity virtual verification environment to eliminate steady-state position sag.

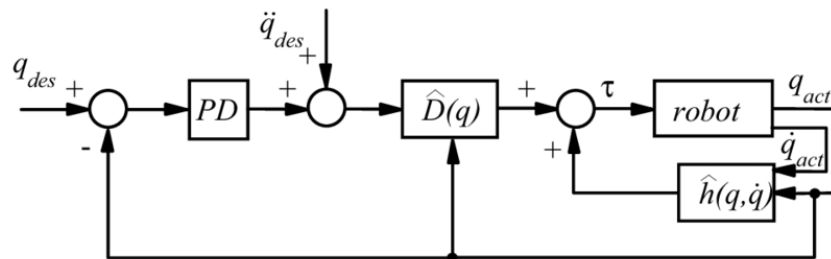


Figure 1 A model based CTC for robotic arm

Computed Torque Control (CTC) is a control method that uses the robot's physics equations to calculate the exact motor force needed to follow a path perfectly.

It cancels out the robot's weight, inertia, and friction so the arm moves exactly like a weightless particle in space.

Chapter 2 : Kinematic and Dynamic Modeling Foundations

To construct a comprehensive mathematical model of a multi-degree-of-freedom manipulator, a structured sequence of kinematic and dynamic formulations must be systematically established.

2.1 Principles of Forward Kinematics

Forward kinematics defines the analytical procedure used to calculate the absolute position and orientation of the robot's end-effector in Cartesian operational space, given a specific vector of joint displacements. This geometric mapping relies entirely on a standardized kinematic framework to ensure that calculations remain consistent and mathematically sound across all six degrees of freedom.

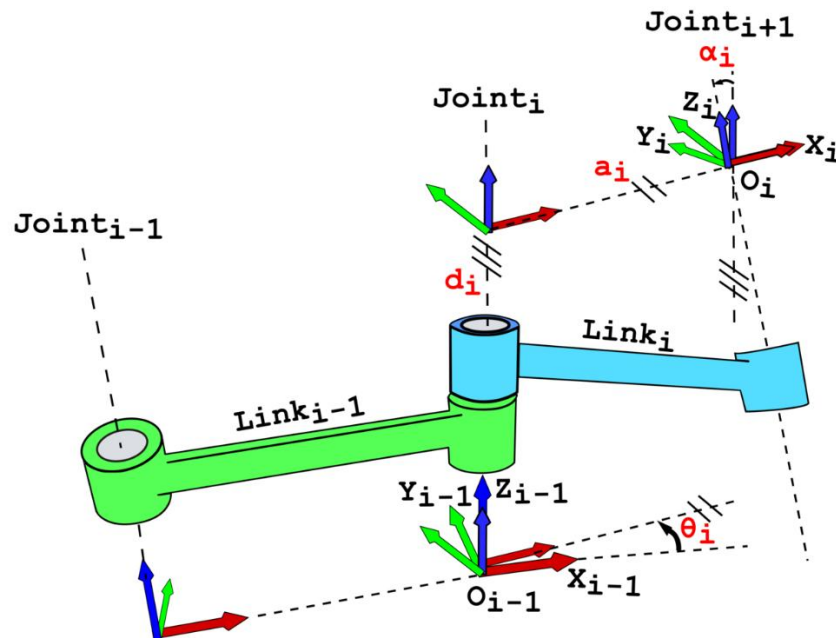


Figure 2 distal approach of DH parameters frames

2.2 Denavit-Hartenberg (DH) Parameterization

To achieve this geometric description, the system utilizes **Denavit-Hartenberg (DH) parameters**. The DH convention simplifies the spatial geometry of a serial manipulator by reducing the relative transformations between adjacent links to exactly four fundamental parameters:

- **Link length** (a_n)
- **Link twist** (α_n)
- **Link offset** (d_n)

- **Joint angle (θ_n)**

Together, these parameters provide the complete, unambiguous geometric blueprint required to construct the robot's forward kinematic model.

2.3 Comparative Conventions: Proximal vs. Distal Approaches

In modern robotics literature, there are two primary methods used to establish and assign these coordinate frame parameters: the **classic (distal) approach** and the **modified (proximal) approach**. The classic approach attaches the reference coordinate frame (n) to the distal end of link (n), meaning the joint axis (n) is aligned with the origin of frame (n-1). In contrast, the modified proximal approach attaches coordinate frame (n) to the proximal joint axis (n), meaning the joint axis (n) aligns directly with the origin of frame (n).

$$\begin{aligned} \text{Trans}_{z_{n-1}}(d_n) &= \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_n \\ \hline 0 & 0 & 0 & 1 \end{array} \right] \\ \text{Rot}_{z_{n-1}}(\theta_n) &= \left[\begin{array}{ccc|c} \cos \theta_n & -\sin \theta_n & 0 & 0 \\ \sin \theta_n & \cos \theta_n & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array} \right] \\ \text{Trans}_{x_n}(r_n) &= \left[\begin{array}{ccc|c} 1 & 0 & 0 & r_n \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array} \right] \\ \text{Rot}_{x_n}(\alpha_n) &= \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha_n & -\sin \alpha_n & 0 \\ 0 & \sin \alpha_n & \cos \alpha_n & 0 \\ \hline 0 & 0 & 0 & 1 \end{array} \right] \end{aligned}$$

2.4 Selection of the Proximal Approach for Dynamic Modeling

For the scope of this parameter estimation work, the **modified proximal DH convention** is explicitly selected. While both approaches yield identical end-effector positioning results, the proximal convention is highly preferred for dynamic identification frameworks like the BIRDy benchmark. By aligning the coordinate frame directly with the proximal joint axis, the resulting Newton-Euler dynamic equations and structural regressor matrices become significantly more streamlined and computationally efficient to evaluate in MATLAB.

2.5 Commutative Properties of Axis-Aligned Transformations

In general spatial kinematics, matrix multiplication is non-commutative, meaning that changing the sequence of geometric operations alters the final spatial result. However, a unique mathematical exception occurs during the derivation of Denavit-Hartenberg transformation chains. For a pair of consecutive rotation and translation operations that share the exact same coordinate axis, the order of matrix multiplication does not affect the outcome, allowing the sequence of transformations to be reversed without altering the final matrix.

2.6 Mathematical Justification via Shared Spatial Axes

This commutative behavior is strictly possible because the translation and rotation transformations operate natively along and around identical coordinate axes. For example, a translation along the (Z)-axis commutes perfectly with a rotation around the (Z)-axis, just as an (X)-axis translation commutes with an (X)-axis rotation. Because these specific operator pairs commute, their algebraic product remains identical regardless of whether the translation or the rotation is executed first.

2.7 Kinematic Implications and Example Formulation

This structural property significantly simplifies the expansion of symbolic kinematic equations within the proximal DH framework. It provides flexibility when compounding local link transformations, as the ordering of internal axis-aligned matrix pairs can be swapped to optimize algebraic reduction. For example, the consecutive transformation sequence can be explicitly demonstrated as follows:

$$(T_{Rz}(\theta) \cdot T_{Rz}(d)) = (T_{Rz}(\theta) \cdot T_{Rz}(d)).$$

$$(T_{Rx}(\alpha) \cdot T_{Dx}(d)) = (T_{Dx}(d) \cdot T_{Rx}(\alpha)).$$

$${}^{n-1}T_n = \text{Trans}_{z_{n-1}}(d_n) \cdot \text{Rot}_{z_{n-1}}(\theta_n) \cdot \text{Trans}_{x_n}(r_n) \cdot \text{Rot}_{x_n}(\alpha_n)$$

$${}^{n-1}T_n = \text{Rot}_{z_{n-1}}(\theta_n) \cdot \text{Trans}_{z_{n-1}}(d_n) \cdot \text{Trans}_{x_n}(r_n) \cdot \text{Rot}_{x_n}(\alpha_n)$$

$${}^{n-1}T_n = \left[\begin{array}{ccc|c} \cos \theta_n & -\sin \theta_n \cos \alpha_n & \sin \theta_n \sin \alpha_n & r_n \cos \theta_n \\ \sin \theta_n & \cos \theta_n \cos \alpha_n & -\cos \theta_n \sin \alpha_n & r_n \sin \theta_n \\ 0 & \sin \alpha_n & \cos \alpha_n & d_n \\ \hline 0 & 0 & 0 & 1 \end{array} \right] = \left[\begin{array}{ccc|c} & & & \\ & R & & T \\ \hline 0 & 0 & 0 & 1 \end{array} \right]$$

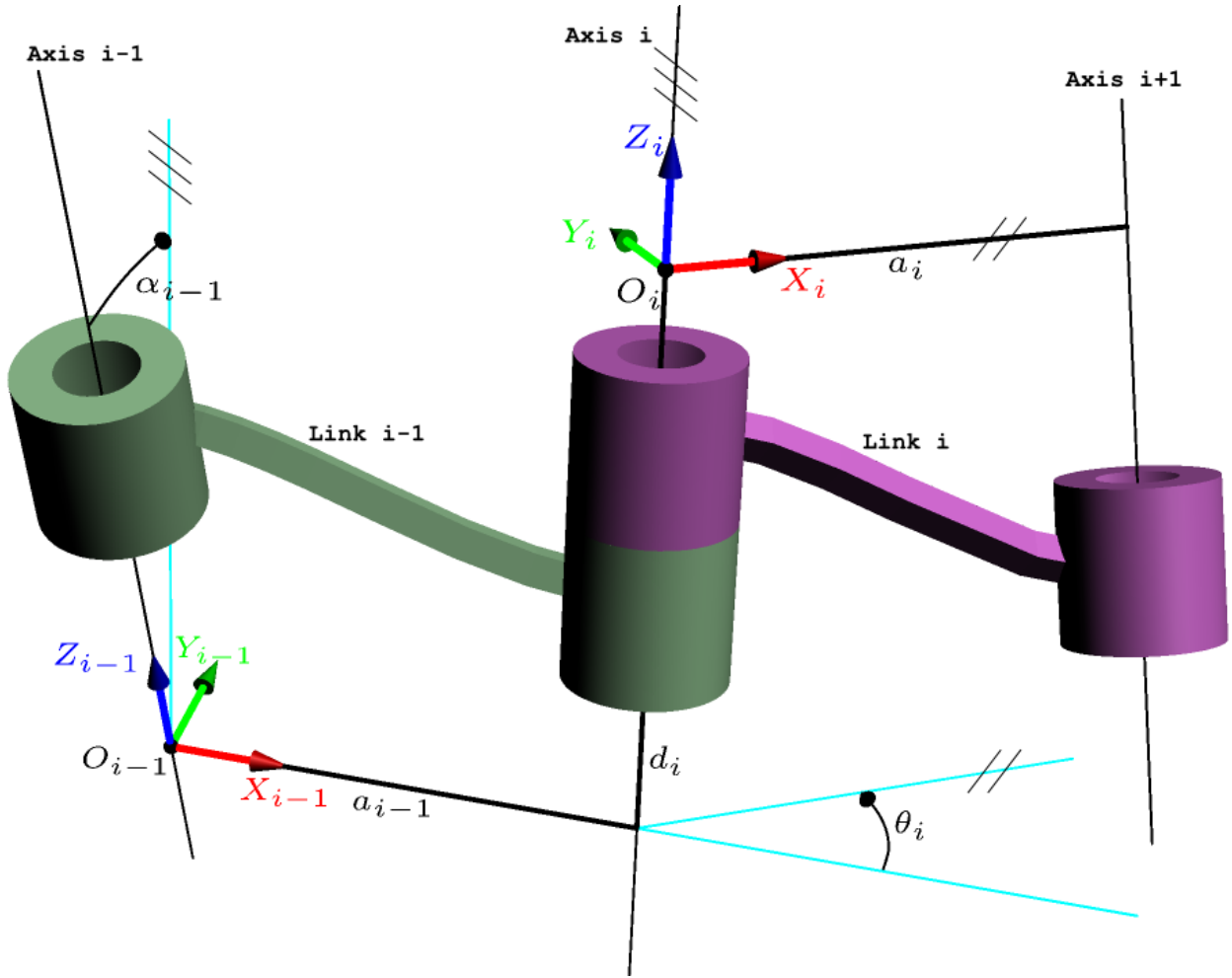


Figure 3 the proximal approach of DH parameters frames

Compared with the classic DH parameters, the coordinates of frame are put on axis $i - 1$, not the axis i in classic DH convention. The coordinates of is put on the axis i , not the axis $i + 1$ in classic DH convention. Another difference is that according to the modified convention,

The transform matrix is given by the following order of operations:

$${}^{n-1}T_n = \text{Rot}_{x_{n-1}}(\alpha_{n-1}) \cdot \text{Trans}_{x_{n-1}}(a_{n-1}) \cdot \text{Rot}_{z_n}(\theta_n) \cdot \text{Trans}_{z_n}(d_n)$$

$${}^{n-1}T_n = \left[\begin{array}{ccc|c} \cos \theta_n & -\sin \theta_n & 0 & a_{n-1} \\ \sin \theta_n \cos \alpha_{n-1} & \cos \theta_n \cos \alpha_{n-1} & -\sin \alpha_{n-1} & -d_n \sin \alpha_{n-1} \\ \sin \theta_n \sin \alpha_{n-1} & \cos \theta_n \sin \alpha_{n-1} & \cos \alpha_{n-1} & d_n \cos \alpha_{n-1} \\ \hline 0 & 0 & 0 & 1 \end{array} \right]$$

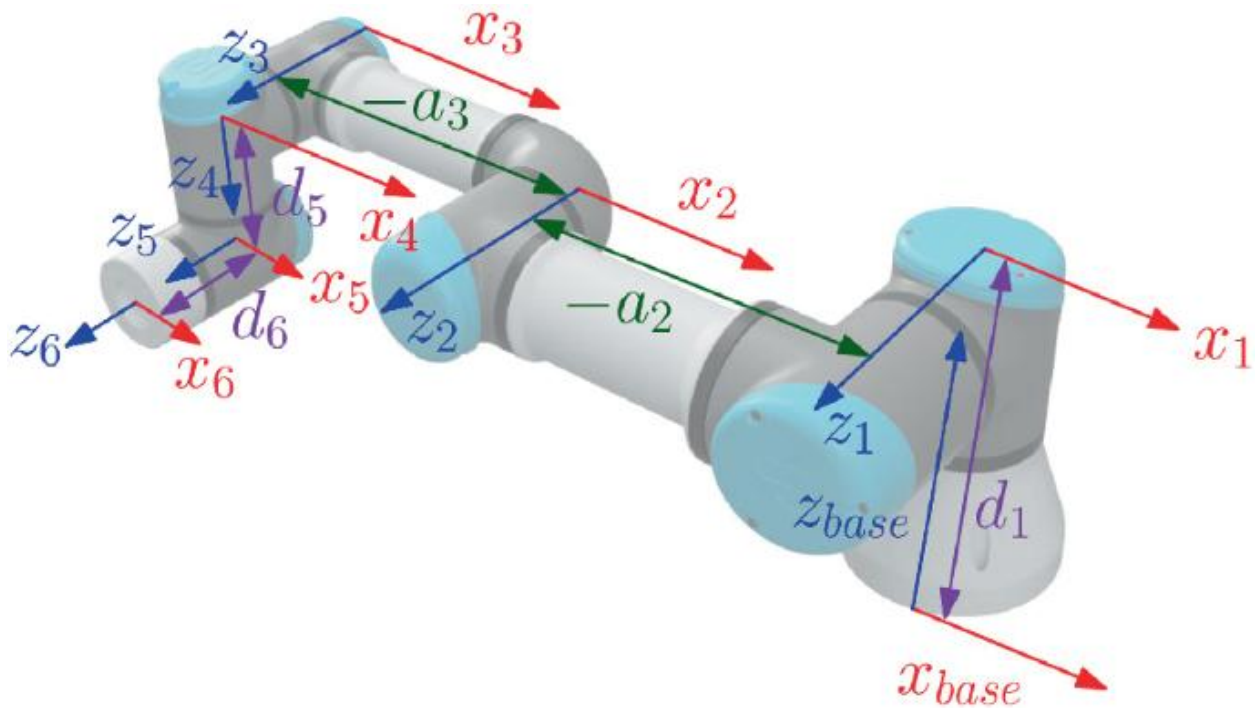


Figure 4 distal frames of the case study robot UR3

Link	r	α	d	θ
1	0	$\frac{\pi}{2}$	d_1	θ_1
2	$-a_2$	0	0	θ_2
3	$-a_3$	0	0	θ_3
4	0	$\frac{\pi}{2}$	d_4	θ_4
5	0	$-\frac{\pi}{2}$	d_5	θ_5
6	0	0	d_6	θ_6

Table 1 distal DH parameters of UR3

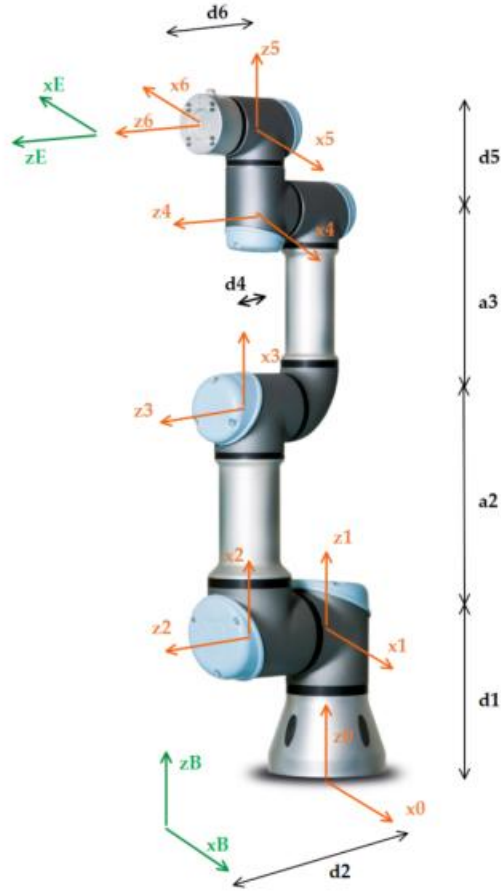


Figure 5 proximal DH parameters frames

Link	$a(i-1)$	$\alpha(i-1)$	d	θ
1	0	0	$d1$	$\theta1$
2	0	$\frac{\pi}{2}$	0	$\theta2$
3	$a2$	0	0	$\theta3$
4	$a3$	0	$-d4$	$\theta4$
5	0	$-\frac{\pi}{2}$	$d5$	$\theta5$
6	0	$\frac{\pi}{2}$	$d6$	$\theta6$

Table 2 proximal DH parameters

2.8 Computational Implementation of Transformation Chains

The systematic assembly of the homogeneous transformation sequences represents the foundational first step of the practical implementation within the MATLAB environment. By executing these matrix cascades at the absolute beginning of the program, the script establishes the complete kinematic blueprint of the 6-DOF arm. This software architecture ensures that all downstream dynamic, inertial, and differential velocity evaluations reference a structurally sound geometric baseline.

2.9 Integration of the Proximal Kinematic Framework

Within this script initialization, the homogeneous transformation matrices are constructed strictly adhering to the modified proximal Denavit-Hartenberg convention. Rather than applying the classic distal frame layout, the frame assignment loops link adjacent coordinate systems directly at the proximal joint axes. The specific engineering advantages and mathematical justifications for prioritizing this proximal approach over the classic method within our identification pipeline.

2.10 The reason of choosing the modified proximal DH convention

The core reason the modified proximal DH convention is explicitly chosen for dynamic parameter estimation boils down to one fundamental rule of physics: It isolates the joint inertial and friction parameters into independent, localized link equations.

2.11 Global Mapping to the World Reference Frame

To establish a rigorous dynamic identification model, every localized coordinate frame along the serial chain must be mapped back to a singular, global coordinate system. This is achieved by compounding the sequential proximal Denavit-Hartenberg matrices to define the homogeneous transformation of each link frame (n) relative to the absolute world reference frame (T_n^0). Without this global baseline, it is mathematically impossible to resolve the net directional vectors of gravitational forces and spatial accelerations acting across the 6-DOF manipulator.

2.12 Defining the Center of Mass (COM) Reference Frames

Beyond tracking the structural joint origins, the full dynamic model requires a distinct set of reference frames to isolate the inertial behaviors of the physical links: the Center of Mass (COM) frames. While joint bearings dictate the kinematic boundaries, the actual kinetic and potential energy of the robot is distributed around the centroid of each link's physical geometry. Defining the precise spatial location of every COM frame relative to the global world frame is a critical prerequisite for building both a high-fidelity simulation model and a valid parameter estimation regressor matrix.

2.13 Practical Utility in System Identification

In an uncalibrated legacy robot, the local offset coordinates (X_n, Y_n, Z_n) are entirely unknown parameters that must be estimated. By structuring the COM frame transformations through this explicit post-multiplication step, the resulting symbolic torque equations maintain their essential linear-in-the-parameters configuration. This allows the unknown center of mass properties to be grouped directly with the link mass as identifiable first-moment variables (MX_n, MY_n, MZ_n), which are then solved cleanly using your MATLAB regression script.

2.14 A deep dive defining the Center of Mass

Transformation to every link to the world frame (defining every link to the world frame) we need to define the center of mass frame of every link to the world frame this step is needed in creating a full detailed dynamics and to

create identification model, $\begin{bmatrix} 1 & 0 & 0 & X_n \\ 0 & 1 & 0 & Y_n \\ 0 & 0 & 1 & Z_n \\ 0 & 0 & 0 & 1 \end{bmatrix}$ by post multiplying this matrix to the homogenous transformation of

any center of mass of any link and also to provide all first moments of our link dynamics, $\begin{bmatrix} 1 & 0 & 0 & MnX_n \\ 0 & 1 & 0 & MnY_n \\ 0 & 0 & 1 & MnZ_n \\ 0 & 0 & 0 & 1 \end{bmatrix}$,

M_n : the mass of the link , X_n : the x coordinate of center of mass of the link,

Y_n : y coordinate of the center of mass aof thelink, Z_n : z coordinate of center ofmass of the link

3. Core Principles of Differential Kinematics

Differential kinematics establishes the analytical relationship between the velocities of the robot joints in the joint space and the resulting spatial velocities of a given link frame or the end-effector in the task space. While forward kinematics maps static joint displacements to absolute Cartesian coordinates, differential kinematics analyzes the

system in motion. This linear mapping is a fundamental prerequisite for dynamic identification, trajectory tracking, and model-based controller design.

	revolute i -th joint
$J_{Li}(q)\dot{q}_i$	$(z_{i-1} \times p_{i-1,E})\dot{\theta}_i$
$J_{Ai}(q)\dot{q}_i$	$z_{i-1}\dot{\theta}_i$

Table 3

Z_{i-1} : The third column of each homogenous transformation matrix of the link pre multiplied by rotation component of the homogenous transformation of the link defined to world frame

$P_{i-1,E}$: The difference of the first three elements of the fourth column of homogenous transformation matrix of the joint from first three elements of the fourth column of homogenous of the last joint defined to the world frame

To bridge the gap between kinematic movement and physical force propagation, the differential model requires two distinct types of velocity mappings: the **Joint Jacobian** and the **Center of Mass (COM) Jacobian**. While the standard Joint Jacobian computes the spatial velocities at the structural joint origins for kinematic path tracking, the COM Jacobian maps joint velocities directly to the volumetric centroid of each individual link. Isolating both matrices is mandatory because mass distributions—and the resulting inertial torques—accelerate around the center of mass rather than the joint axes.

4. The Primacy of the Symbolic Formulation

When modeling a 6-Degree-of-Freedom manipulator for parameter estimation, the specific choice of dynamic derivation algorithm is secondary to a more critical structural requirement: obtaining the final equations of motion in an explicit **symbolic form**. Numerical methods can compute the net torque requirements for a single instance in time, but they treat the underlying system properties as a hidden black box. A symbolic formulation maps the nonlinear configurations of the robot directly to its physical attributes, which is a mandatory prerequisite for isolating individual link properties.

$$\tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) + F(\dot{q}) \quad (1)$$

τ : The torque vector $R^{6 \times 1}$, $M(q)$: The inertia matrix $R^{6 \times 6}$, $C(q, \dot{q})$: Coriolis and centrifugal force matrix $R^{6 \times 6}$,

$G(q)$: The gravity vector $R^{6 \times 1}$ $F(\dot{q})$: is two diagonal matrices one describe coulumb friction torque and the other viscous friction torque $R^{6 \times 6}$.

5. Summary of Design Objectives

The core intent of the preceding analytical formulations is not to provide a redundant or exhaustive textbook explanation of universal robot kinematics and dynamics. Rather, these derivations have been systematically detailed to provide the explicit mathematical justification for why specific modeling methodologies were prioritized over alternative academic frameworks for this 6-DOF benchmark. Every kinematic mapping, matrix post-multiplication,

and dual-Jacobian expansion served a deliberate strategic purpose: establishing a highly optimized, linear, and computationally sound baseline for the downstream parameter identification pipeline.

By reviewing the structural layout of the equations, the engineering advantages of these selections become clear:

- **The Proximal DH Convention** was chosen over the classic distal approach specifically to cleanly isolate joint friction and rotor inertia parameters into localized link matrices.
- **The Matrix Post-Multiplication Step** was executed to gracefully transform nonlinear center-of-mass coordinates into linear first-moment variables (MX, MY, MZ), rendering the estimation problem solvable via linear regression.
- **The Dual-Jacobian Framework** was derived to map spatial accelerations directly to the link centroids, matching the strict recursive boundaries of the symbolic Newton-Euler algorithm.

Ultimately, this mathematical architecture serves as the mandatory bridge linking our theoretical 6-DOF model to the physical realities of the Simscape Multibody environment. With the symbolic equations successfully structured to maintain a **linear-in-the-parameters** configuration, the framework is fully prepared for numerical execution. The subsequent chapter will detail how this symbolic blueprint is deployed within MATLAB to filter raw sensor data, resolve rank deficiencies via base parameter reduction, and execute the final parameter estimation algorithms.

Inertial parameter estimation

1.Introduction

1.1Academic Context and Control Challenges

The expanding deployment of advanced adaptive, predictive, and passivity-based control strategies in modern robotic applications presents significant structural challenges for researchers and engineers. Because these high-performance control architectures rely strictly on the explicit mathematical formulation of the inverse system dynamics, their baseline accuracy is inherently limited by the quality of the internal plant model. Consequently, substantial research efforts are focused on the field of inertial parameter identification to optimize tracking precision, reject disturbances, and maximize overall controller robustness.

1.2 Mechanics of Dynamic Parameter Identification

In the context of rigid multi-body robot manipulators, classical frameworks for identifying inertial parameters rely on a rigorous analysis of joint kinematics and torque profiles captured over a distinct, finite time horizon. The identification pipeline requires the manipulator to track a specialized excitation trajectory designed to sweep the full velocity and acceleration boundaries of its dynamic components. The resulting residuals—representing the deviations between measured physical states and estimated mathematical outputs—are then fed back into optimization loops to iteratively refine the parameter estimates.

1.3 Preeminence of the Linear Least-Squares Framework

In the context of rigid multi-body robot manipulators, classical frameworks for identifying inertial parameters rely on a rigorous analysis of joint kinematics and torque profiles captured over a distinct, finite time horizon. The identification pipeline requires the manipulator to track a specialized excitation trajectory designed to sweep the full velocity and acceleration boundaries of its dynamic components. The resulting residuals—representing the deviations between measured physical states and estimated mathematical outputs—are then fed back into optimization loops to iteratively refine the parameter estimates.

1.4 Simulative Validation vs. Real-World Physical Constraints

In practical applications, evaluating the performance of an identification algorithm on an actual physical system presents severe challenges due to unavoidable modeling complexities. Real-world manipulators exhibit a wide variety of complex, high-order physical phenomena—such as gear backlash, link flexibility, and highly non-linear structural friction—that are exceptionally difficult to capture mathematically. Consequently, simulative validation serves as a vital benchmark for performance quantification. By utilizing a high-fidelity simulated environment, the ground-truth parameters are fully known, which enables a decoupled analysis of how isolated disturbances, such as measurement noise or specific friction profiles, influence estimate convergence and quality.

1.5 Design and Optimization of Exciting Trajectories

To guarantee the accurate numerical convergence of either the base parameters(β) or the full standard parameter vector (x) the robot must execute a highly exciting trajectory. If the joint motions are too simple—such as moving only one joint at a time or moving at constant velocities—the inertial, centrifugal, and Coriolis forces will not physically manifest. This results in an ill-conditioned observation regressor matrix where key dynamic columns drop to zero, hiding vital parameters from the estimator solver.

To maximize parameter observability, the excitation trajectory is mathematically optimized using three core criteria:

Harmonic Frequency Content: The trajectory is typically parameterized using finite Fourier series waves for each joint. This multi-frequency approach ensures that positions, velocities, and accelerations vary continuously and simultaneously across all degrees of freedom. [

Condition Number Minimization: The optimal kinematic parameters (frequencies and amplitudes) are found by minimizing the condition number of the tracking observation matrix (W). A low condition number guarantees that the matrix is well-conditioned and highly immune to measurement noise.

Physical Boundary Constraints: The optimization algorithm ensures that the generated trajectories strictly respect the physical limits of the robot actuators. This keeps joint positions, maximum velocities, and peak torques safely within the safe operational limits of the hardware.

2. Inverse dynamics model and the linearity between the torque and inertial parameters(standard parameters)

2.1 The standard parameter and the origin of the parameter estimation

The inverse dynamic model (IDM) of a rigid serial robot manipulator with n -degrees of-freedom relates the joint-space motion quantities $[q, \dot{q}, \ddot{q}] \in R^n$, to the generalized forces $\tau_{idm} \in R^n$ applied to the system. The IDM can be derived using the Euler-Lagrange formalism

Where $M(\chi, q) \in R^{n \times n}$ denotes the generalized inertia matrix, $C(\chi, q, \dot{q}) \in R^{n \times n}$ the Coriolis and centripetal effects matrix, $g(\chi, q) \in R^n$ the gravitational torque vector and $\zeta(\chi, \dot{q}) \in R^n$ the friction vector. This equation is parameterized by the vector $\chi = [\chi_1 \chi_2 \cdots \chi_n] \in R^p$ concatenating the standard dynamic parameters of each robot link j , expressed as

$$\chi_j = [XX_j, XY_j, XZ_j, YY_j, YZ_j, ZZ_j, MX_j, MY_j, MZ_j, M_j, I_{aj}, F_{vj}, F_{cj}]^T$$

By contrast, the IDM has the interesting property of being linear in χ , and can thus be reformulated as $\tau_{idm} = Y\chi(q, \dot{q}, \ddot{q})$

Where $Y\chi(q, \dot{q}, \ddot{q}) = \frac{\partial \tau_{idm}}{\partial \chi} \in R^{n \times p}$ (2)

Denotes the closed-form expression of the Jacobian matrix of τ_{idm} with respect to χ , often referred to as the model regressor. It is worth noting that the equality in Equation (2) only holds provided that the friction term $\zeta(\chi, \dot{q})$ is also linear with respect to the parameter vector

so we choose the only viscous and coulumb friction model in our work cause they are the only ones with linearity with respect to parameter vector

Under these assumptions, dynamic parameters identification can be considered to be an inverse problem. Sampling (2) at multiple different time epochs along a given state trajectory results in an over-determined observation system of linear equations in χ , for which a unique solution can be derived provided that the system is not rank deficient. Note however that this latter point turns out to be hardly verifiable in practice

2.2 The rank of sampled observation

The rank of the sampled observation system is function of two factors, namely the nature of the collected data samples, and the internal structure of the regressor matrix $Y\chi$. Data rank deficiency of the observation system is a direct consequence of noisy or improper data samples that do not adequately excite the model parameters (i.e., that do not sufficiently reveal their influence on the system dynamics). This issue can most of the time be solved by having the robot track a trajectory that is specifically designed to excite the model parameters

The other aspect of the problem lies in the fact that the intrinsic geometrical properties of robot manipulators naturally imply that some of the dynamic parameters within the χ vector simply cannot be identified, for having rigorously no influence on the actual robot motions whatever the followed trajectory

In the same manner, some parameters can only be identified jointly because of their combined influence on the system. From a formal point of view, these issues can be characterized as a structural rank deficiency of the regression matrix $Y\chi$.

To solve this issue, it is necessary to perform a set of column rearrangements within $Y\chi$, eventually leading to the set of $b \leq p$ base parameters that are actually linear combinations of the standard parameters χ .

2.3 Base parameter and identification model :

Base inertial parameters are defined as the set of dynamic parameters which is sufficient to completely describe the— intrinsically constrained—dynamics of a robot mechanism.

The base parameter vector $\beta \in R^b$ can be obtained by performing a set of rearrangements in the symbolic expressions of equation (2) (**note that in this context, some of the performed symbolic simplifications are only made possible using proximal DH convention**).

Dedicated numerical methods based QR decomposition will help us obtaining both base parameter β and the related symbolic regressor model Y_β .

3. Constructing the standard parameter regressor model $Y\chi$

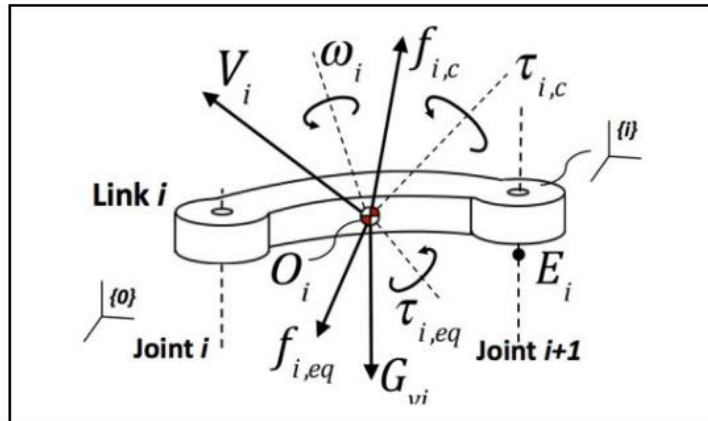


Figure 6 Free-body diagram of link i of a robot manipulator

3.1 Linear and Angular Momentum Formulations of an Intermediate Link

Consider an intermediate link i of an n -DOF robot manipulator as shown in Figure 6, where $f_{i,eq}$ and $t_{i,eq}$ are the equivalent external force and moment exerted at the center of mass O_i of the body, respectively. The linear momentum of link i is

$$n_{vi} = m_i v_i = m_i j_{vi} \dot{q} \quad (3)$$

Where m_i and v_i , respectively, denote the mass and the linear velocity at O_i , and $j_{vi} \in \mathbb{R}^{3 \times n}$ is the kinematic mapping from joint velocity $\dot{q}$ to v_i . Similarly, the angular momentum of link i is

$$n_{wi} = I_i \omega_i = R_i I_i^i R_i^T j_{wi} \dot{q} \quad (4)$$

where ω_i is the angular velocity of link i ; $I_i \in \mathbb{R}^{3 \times 3}$ denote the inertia tensors of link i at O_i w.r.t. the reference frame, $\{0\}$, and w.r.t. the body frame, $\{i\}$, respectively; $R_i I_i^i R_i^T$ is the rotation matrix from frame $\{0\}$, to frame $\{i\}$; and $j_{wi} \in \mathbb{R}^{3 \times n}$ the kinematic map from joint velocity $\dot{q}$ to ω_i . Note that I_i^i is a constant matrix, whereas I_i is varying with the configuration of the robot.

Now, taking time derivatives on equations (3) and (4) yields

$$\begin{bmatrix} \dot{v}_{vi} \\ \dot{\omega}_{wi} \end{bmatrix} = - \begin{bmatrix} \mathbf{G}_{vi} \\ \mathbf{0} \end{bmatrix} + \begin{bmatrix} \mathbf{f}_{i,c} \\ \boldsymbol{\tau}_{i,c} \end{bmatrix} + \begin{bmatrix} \mathbf{f}_{i,eq} \\ \boldsymbol{\tau}_{i,eq} \end{bmatrix}$$

Where $\mathbf{G}_{vi} := m_i \mathbf{g}_r$ denotes the gravitational force with $\mathbf{g}_r$ being the gravitational acceleration vector, and $\mathbf{f}_{i,c}$ and $\boldsymbol{\tau}_{i,c}$, respectively, are the equivalent constraint force and moment acting on O_i . We can combine the above relation over the entire system as

$$\dot{n} = -G + F_c + F \quad (4)$$

Where :-

$$\begin{aligned} \nu &= [\nu_{v,1}^T \nu_{v,2}^T \dots \nu_{v,n}^T | \nu_{w,1}^T \nu_{w,2}^T \dots \nu_{w,n}^T]^T \\ \mathbf{G} &= [\mathbf{G}_{v,1}^T \mathbf{G}_{v,2}^T \dots \mathbf{G}_{v,n}^T | \mathbf{0}^T \mathbf{0}^T \dots \mathbf{0}^T]^T \\ \mathbf{F}_c &= [\mathbf{f}_{c,1}^T \mathbf{f}_{c,2}^T \dots \mathbf{f}_{c,n}^T | \boldsymbol{\tau}_{c,1}^T \boldsymbol{\tau}_{c,2}^T \dots \boldsymbol{\tau}_{c,n}^T]^T \\ \mathbf{F} &= [\mathbf{f}_{1,eq}^T \mathbf{f}_{2,eq}^T \dots \mathbf{f}_{n,eq}^T | \boldsymbol{\tau}_{1,eq}^T \boldsymbol{\tau}_{2,eq}^T \dots \boldsymbol{\tau}_{n,eq}^T]^T \end{aligned}$$

The momentum on other hand, can be cast into the following form $n = M(q) * J(q) * \dot{q}$ (5)

where

$$\begin{aligned} \mathcal{M}(q) &= \text{blockdiag}[m_1 \mathbf{I}_{3 \times 3} \dots m_n \mathbf{I}_{3 \times 3} | \mathbb{I}_1 \dots \mathbb{I}_n] \\ \mathbf{J}(q) &= [\mathbf{J}_{v1}^T \mathbf{J}_{v2}^T \dots \mathbf{J}_{vn}^T | \mathbf{J}_{w1}^T \mathbf{J}_{w2}^T \dots \mathbf{J}_{wn}^T]^T \end{aligned}$$

With $\mathbf{I}_{3 \times 3}$ being the 3×3 identity matrix. Plugging equation (5) into equation (4) and then pre-multiplying $J^T(q)$ yield

$$J^T M J \ddot{q} + (J^T \dot{M} J + J^T M \dot{J}) \dot{q} = J^T G + J^T F_c + J^T F \quad (6)$$

Where we omitted the dependency of q in J and M for simplicity. Since the virtual work principle gives

$$\tau = J^T F \text{ and } J^T F c = 0$$

where τ is the joint torque, equation (6) becomes

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau \quad (7)$$

$$M = J^T M J, C = J^T \dot{M} J + J^T M \dot{J}; \text{ and } g = J^T G$$

If written in a summation form, equation (7) finally becomes

$$\begin{aligned} & \sum_{i=1}^n (m_i J_{vi}^T J_{vi} + J_{wi}^T R_i \mathbb{I}_i^T R_i^T J_{wi}) \ddot{q} + \sum_{i=1}^n (m_i J_{vi}^T \dot{J}_{vi} \\ & + J_{wi}^T R_i \mathbb{I}_i^T \dot{R}_i^T J_{wi} + J_{wi}^T S(\omega_i) R_i \mathbb{I}_i^T R_i^T J_{wi}) \dot{q} \\ & + \sum_{i=1}^n J_{vi}^T G_{vi} = \tau \end{aligned} \quad (8)$$

Where $S(\omega i) := (\omega i \times)$ is a matrix operator of vector cross product.

3.2 Eliminating the unknown COM O_i of the equation

In spite of its elegance, equation (8) is not useful in its current form to be cast into a parameter regression form, because J_{vi} 's (and their derivatives) are not computable without prior knowledge of O_i , the center of mass of link i , which is unknown before identification. In the next subsection, equation (8) shall be modified by translating the point of interest from O_i to a known kinematic position to yield an effective dynamic equation giving us a convenient regression model.

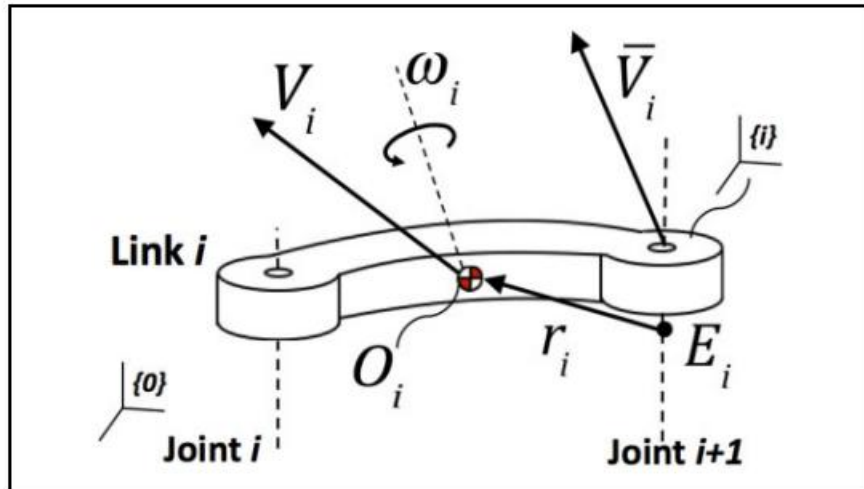


Figure 7 Velocity of link i of a robot manipulator in translated coordinate.

Suppose $\bar{V}_i$ be the linear velocity at E_i , the known distal end of link i , and also suppose r_i be the distance vector from E_i to O_i (refer to Figure 7). Then, it is true that

$$V_i = \bar{V}_i + \omega_i \times r_i = (\bar{J}_{vi} - S(r_i) * J_{wi}) * \dot{q} \quad (9)$$

Where $\bar{J}_{vi}$ is defined such that $\bar{V}_i = \bar{J}_{vi} * \dot{q}$. Thus, we obtain a Jacobian relation as

$$J_{vi} = \bar{J}_{vi} - S(r_i) * J_{wi} \quad (10)$$

Plugging the above two relations into equation (4) and pre-multiplying J^T yield

$$\begin{aligned} & \sum_{i=1}^n m_i \left(\bar{J}_{vi} - S(r_i) J_{wi} \right)^T (\dot{\bar{V}}_i + \dot{\omega}_i \times r_i + \omega_i \times \dot{r}_i) \\ & + \sum_{i=1}^n J_{wi}^T (\mathbb{I}_i \dot{\omega}_i + \dot{\mathbb{I}}_i \omega_i) + \sum_{i=1}^n \left(\bar{J}_{vi}^T - S(r_i) J_{wi}^T \right)^T (m_i g_r) = \tau. \end{aligned} \quad (11)$$

After expanding and doing mathematical manipulation, the above equation reduces to

$$\begin{aligned} & \sum_{i=1}^n m_i \bar{J}_{vi}^T \dot{\bar{V}}_i + \sum_{i=1}^n m_i \left(\bar{J}_{vi}^T S(\dot{\omega}_i) + \bar{J}_{vi}^T S(\omega_i) S(\omega_i) \right. \\ & \left. - J_{wi}^T S(\dot{\bar{V}}_i) \right) r_i + \sum_{i=1}^n J_{wi}^T \left(R_i^T \mathbb{I}_i R_i^T \dot{\omega}_i + S(\omega_i) R_i^T \mathbb{I}_i R_i^T \omega_i \right) \\ & + \sum_{i=1}^n (\bar{J}_{vi}^T - S(r_i) J_{wi}^T)^T (m_i g_r) = \tau \end{aligned} \quad (12)$$

Where ${}^i I_t := I_i^i - m_i * S(r_i^i) * S(r_i^i)$ means the invariant inertia tensor computed at the translated coordinate, E_i (this is called as Steiner's theorem). Also, note that r_i^i means r_i represented in the body frame $\{i\}$

To derive a linear regression equation from equation (12), we rearrange the parameters of ${}^i I_t$, by following the way

$$\begin{bmatrix} tI_{xx}^i & tI_{xy}^i & tI_{xz}^i \\ tI_{yx}^i & tI_{yy}^i & tI_{yz}^i \\ tI_{zx}^i & tI_{zy}^i & tI_{zz}^i \end{bmatrix} * \begin{bmatrix} d1 \\ d2 \\ d3 \end{bmatrix} = \begin{bmatrix} d1 & 0 & 0 & -d2 & -d3 & 0 \\ 0 & d2 & 0 & -d1 & 0 & -d3 \\ 0 & 0 & d3 & 0 & -d1 & -d2 \end{bmatrix} * \begin{bmatrix} tI_{xx}^i \\ tI_{yy}^i \\ tI_{zz}^i \\ tI_{xy}^i \\ tI_{xz}^i \\ tI_{yz}^i \end{bmatrix}$$

The red marked matrix is $B(d)$ translate the inertia symmetric matrix to vector of $R^{6 \times 1}$

Applying the above transformation to equation (12) yields

$$\begin{aligned}
& \sum_{i=1}^n m_i \bar{\mathbf{J}}_{vi}^T (\dot{\mathbf{V}}_i + \mathbf{g}_r) + \sum_{i=1}^n m_i (\bar{\mathbf{J}}_{vi}^T \mathbf{S}(\dot{\boldsymbol{\omega}}_i) + \bar{\mathbf{J}}_{vi}^T \mathbf{S}(\boldsymbol{\omega}_i) \mathbf{S}(\boldsymbol{\omega}_i) \\
& - \mathbf{J}_{wi}^T \mathbf{S}(\dot{\mathbf{V}}_i + \mathbf{g}_r)) \mathbf{R}_i \mathbf{r}_i \sum_{i=1}^n \mathbf{J}_{wi}^T (\mathbf{R}_i \mathbf{B}(\dot{\boldsymbol{\omega}}_i^i) \\
& + \mathbf{S}(\boldsymbol{\omega}_i) \mathbf{R}_i \mathbf{B}(\boldsymbol{\omega}_i^i)) \gamma_i = \tau
\end{aligned} \tag{13}$$

where $\dot{\boldsymbol{\omega}}_i = \mathbf{R}_i^T \dot{\boldsymbol{\omega}}_i^i$ is the angular acceleration with respect to frame $\{i\}$. All the terms in the above equation are expressed linearly in the inertial parameters: $m_i r_i^i$, and γ_i .

3.3 The Viscous Coulomb friction forces

$$\tau = \tau_a - \tau_f \tag{14}$$

Where τ_a denotes the pure active joint torque and τ_f is the friction torque in the joints which is commonly modeled as

$$\tau_f = \alpha * \text{sgn}(\dot{q}) + \beta * \dot{q} = \sum_{i=1}^n H_i * \mu_i \tag{15}$$

Where $\alpha = \text{diag}[\alpha_1, \dots, \alpha_n]$ and $\beta = \text{diag}[\beta_1, \dots, \beta_n]$ are the matrices of coefficients of Coulomb's friction and viscous friction, respectively; $\text{sgn}(\dot{q})$ is the component wise vector sign function; $H_i \in R^{n \times 2}$ is a matrix whose j th row is $[\text{sgn}(\dot{q}_i)]$ for $i = j$, but is zero for $j \neq i$; and $u_i = [\alpha_i \beta_i]^T$ is the vector of friction coefficients related to the i th joint.

Ultimately, by combining equations (13) to (15), the regression equation in the full dynamic parameters is constructed

$$\sum_{i=1}^n Y_x x = \sum_{i=1}^n [a_i, 1, a_i, 2, a_i, 3, a_i, 4] \begin{bmatrix} m_i \\ m_i r_i^i \\ \gamma_i \\ u_i \end{bmatrix} = \tau \tag{16}$$

The regression equation (16) can be cast into a single big-sized matrix equation as

$$Y_x x = \tau \tag{17}$$

Y_x matrix will be block upper triangular matrix

$$\begin{bmatrix}
* & * & \dots & * & * \\
\mathbf{0} & * & \dots & * & * \\
\vdots & \vdots & \vdots & \vdots & \vdots \\
\mathbf{0} & \mathbf{0} & \dots & * & * \\
\mathbf{0} & \mathbf{0} & \dots & \mathbf{0} & *
\end{bmatrix}$$

3.4 Base parameter identification model Y_b

Consider the observation matrix $W\chi(q^-, q^+, q) \in R^{(n \cdot N) \times p}$ constructed by stacking the samples of Y_x obtained from a randomly generated set of $N \gg 1$ distinct joint values

$$W_x P = Q \begin{bmatrix} R \\ 0_{(r-b) \times p} \end{bmatrix} \quad (18)$$

where for $r = nN$, $Q \in O(r)$, $R \in R^{b \times p}$ is an upper-triangular matrix of rank $b \leq p$ and $P \in O(p)$ is a permutation matrix chosen – by default – so that the diagonal values of R are arranged in decreasing order.

Denoting by $\bar{P} \in R^{p \times b}$ and $\underline{P} \in R^{p \times (p-b)}$ the first b and last $p - b$ columns of P leads to

$$\begin{bmatrix} W_x \bar{P} & W_x \underline{P} \end{bmatrix} = \begin{bmatrix} \bar{Q} & \underline{Q} \end{bmatrix} \begin{bmatrix} \bar{R} & \underline{R} \\ 0_{(r-b) \times p} \end{bmatrix} = \begin{bmatrix} \bar{Q} \bar{R} & \bar{Q} \underline{R} \end{bmatrix} = \bar{Q} \bar{R} \begin{bmatrix} 1^{b \times b} & \bar{R}^{-1} \underline{R} \end{bmatrix} =$$

$$W_x \bar{P} \begin{bmatrix} 1^{b \times b} & \bar{R}^{-1} \underline{R} \end{bmatrix} \quad (19)$$

Where $\bar{R} \in R^{b \times b}$, $\underline{R} \in R^{b \times (p-b)}$ respectively denote the first b and the last $p - b$ columns of R . Since P is orthogonal, one can write

$$W_x X = \begin{bmatrix} W_x \bar{P} & W_x \underline{P} \end{bmatrix} \begin{bmatrix} \bar{X} \\ \underline{X} \end{bmatrix} \quad (20)$$

With $\bar{X} = \bar{P}^T X$ and $\underline{X} = \underline{P}^T X$, yielding

$$W_x X = W_x \bar{P} \begin{bmatrix} 1^{b \times b} & \bar{R}^{-1} * \underline{R} \end{bmatrix} \begin{bmatrix} \bar{X} \\ \underline{X} \end{bmatrix} = W_x * \bar{P} * \beta \quad (21)$$

$$W_x * \bar{P} = Wb \quad (22)$$

Hence the corresponding non-bijective mapping between the base parameter vector β and the standard parameter vector χ

$$\beta = \begin{bmatrix} \bar{P}^T & \bar{R}^{-1} * \underline{R} * \underline{P}^T \end{bmatrix} X \quad (23)$$

3.5 Regressor Model and Base Parameter Validation via Reverse Engineering

To ensure absolute mathematical accuracy in the symbolic modeling phase, the internal structural architecture of the **BIRDy (Benchmark for Identification of Robot Dynamics)** open-source toolbox was systematically reverse-engineered. This deconstruction was executed with a strict, narrow scope: it was used solely to verify the algebraic correctness of the derived linear-in-the-parameters symbolic regressor matrix and to validate the column-reduction logic that isolates the minimum base parameters.

The reverse-engineered workflow was restricted entirely to the following two core elements:

- **Regressor Matrix Structural Audit:** The custom-derived symbolic kinematic regressor equations were cross-referenced against the benchmark's multi-body formulation to guarantee that every trigonometric cross-coupling term was modeled accurately.
- **Base Parameter Reduction Mapping:** The column-pivoting and linear grouping rules used to condense the rank-deficient standard parameters down to the 48 minimum independent base variables were backward-engineered to confirm that your extraction matrix perfectly mirrors established academic standards.
By limiting the use of the benchmark strictly to these modeling and reduction stages, all subsequent trajectory generation, direct state data extraction, and non-linear optimization constraint loops remain entirely independent custom developments. This targeted reverse-engineering step provides an unshakeable mathematical baseline, proving that the structural equations feeding your controller are completely verified before the virtual validation phase begins.

3.6 Rearranging the observation matrix

We need first to rearranging the samples of each joint torque above each other as example if we have two link robot and we have 100 reads so the observation matrix of torque $\in R^{200 \times 1}$ the first 100 elements is associated with the first link the others with the second link hence the first one hundred row of the Wb will be associated with the first link and the other with the second link with arbitrary

column (base parameters number) we can solve the problem with least square method or weighted least square method and identify the base parameter

4. The experiment of estimating the parameter estimation

The dynamic validation is conducted on a UR3 manipulator model, a six-degree-of-freedom robotic system featuring entirely revolute joints. Because the analytical derivation of the full dynamic equations for a six-axis manipulator yields highly complex, high-order symbolic expressions, the explicit mathematical model is substituted with a high-fidelity Simscape Multibody model. This virtual framework acts as a digital twin, capturing the multi-body physical interactions much more accurately than idealized closed-form equations. [1]

To comprehensively evaluate the robustness and convergence of the parameter estimation pipeline under varying operational conditions, two distinct experimental scenarios are designed:

4.1 Experiment | (Noise-Free Benchmark Constraints)

A high-fidelity simulation configuration designed to establish the core tracking baseline under ideal operational constraints. This scenario utilizes a minimized, structurally independent base parameter matrix (β) to map the robot's dynamics. The experiment pairs directly extracted, completely noise-free joint position, velocity, and acceleration quantities with raw joint torque inputs. By operating within a pure, noise-free domain without the need for external friction filtering blocks, this clean pipeline takes a definitive step further to solve the real problem of establishing clean data boundaries, validating that the optimization routine can isolate mass properties with perfect mathematical precision and zero phase distortions.

Excitation Trajectory Profile: Each joint tracks a unique sinusoidal wave profile.

Temporal and Sampling Setup: The total data collection window spans a duration of (10 seconds). Kinematic and dynamic state variables—comprising joint positions, velocities, accelerations, and electromagnetic motor torques—are captured uniformly at a high-speed sampling interval of ($\Delta t = 0.01$ seconds (100 Hz)).

Proportional Gain Calibration: The proportional gain matrix (Kp) is tuned analytically based on the targeted natural frequency ($W_n = 10 \text{ rad/s}$), satisfying the structural

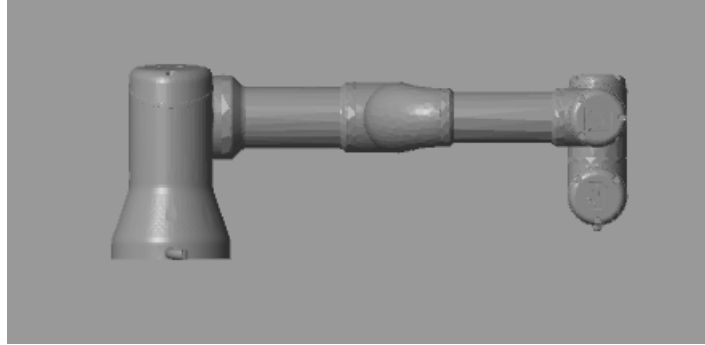


Figure 8 Side view for UR3

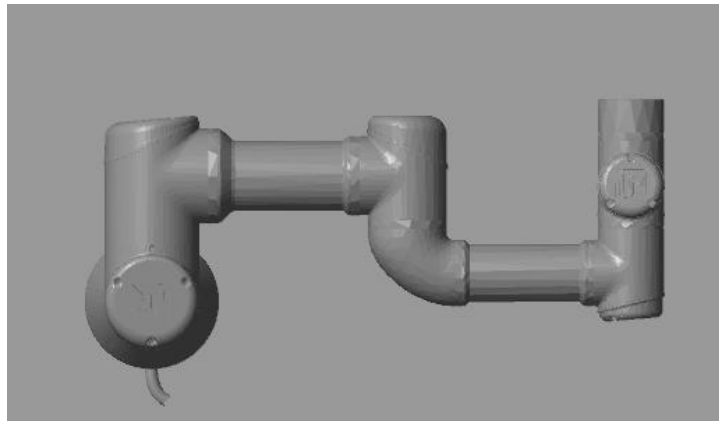


Figure 9 Top view for UR3

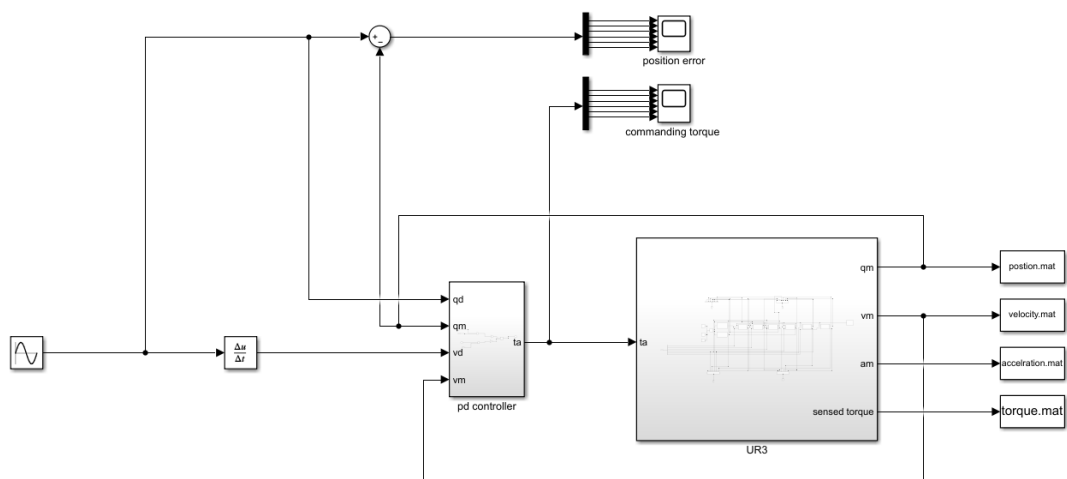


Figure 10 The control loop

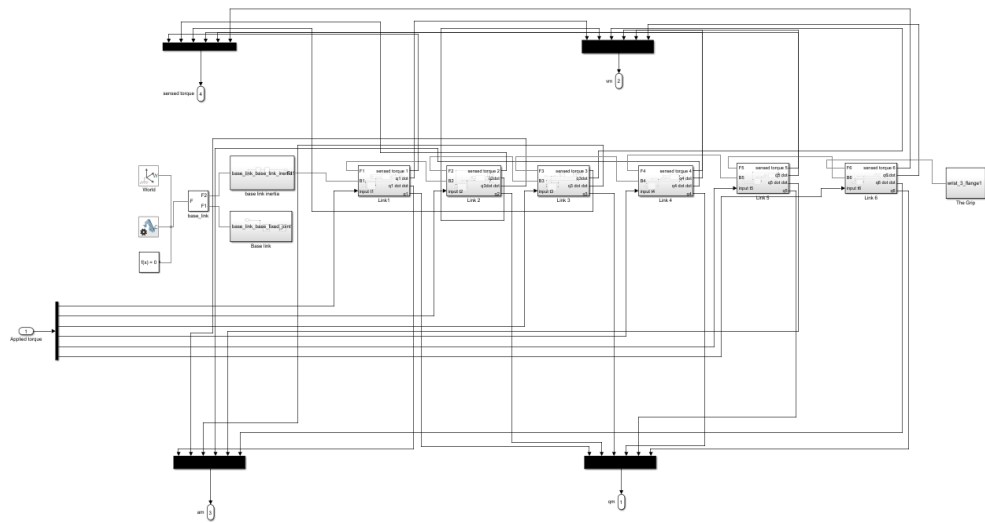


Figure 11 The Simulink Simscape UR3 model

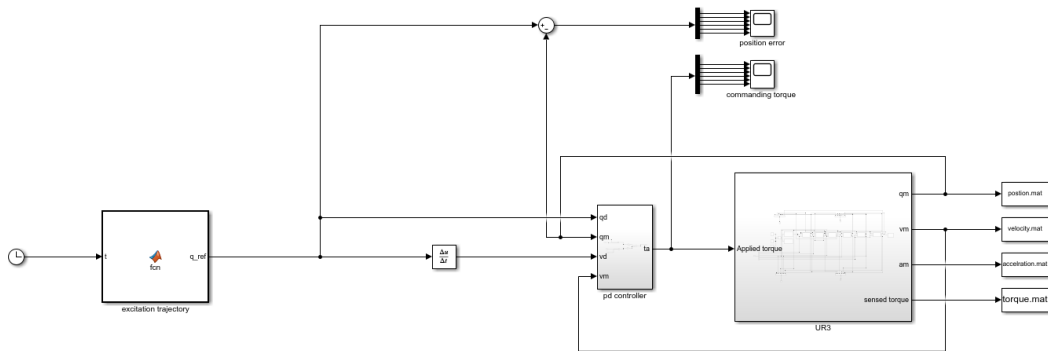


Figure 12 The control loop

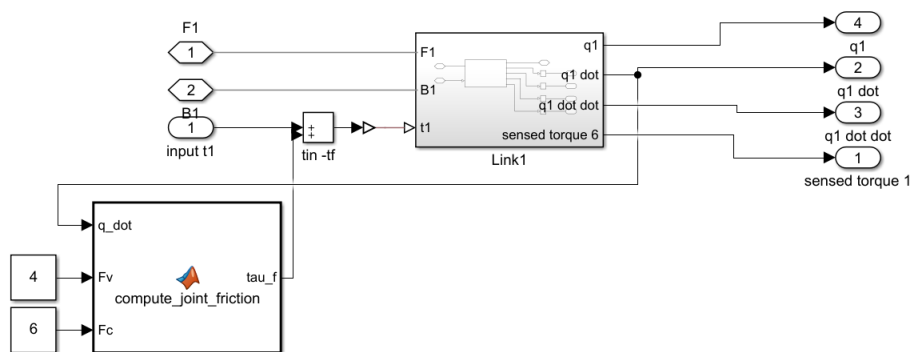


Figure 13 The link wiring with applied friction on

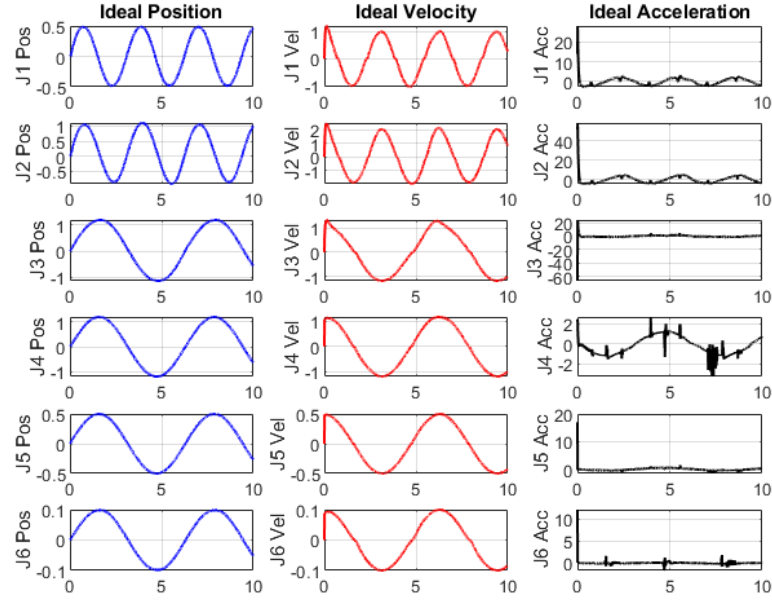


Figure 14 The kinematics of robot arm with friction included

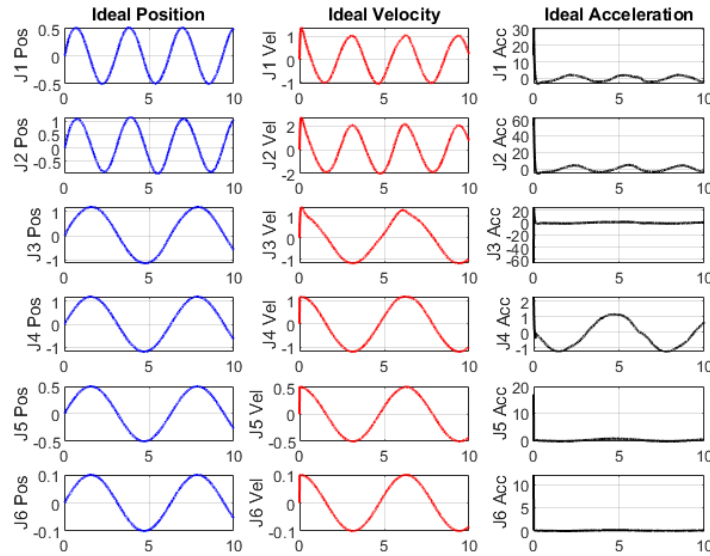


Figure 15 The kinematics of robot arm without friction

During the excitation experiment, the robot arm's state variables exhibit distinct, localized spikes that are amplified within the acceleration plots. These transient spikes provide clear, physical evidence of friction forces acting within the joints, particularly during direction reversals where stick-slip transitions occur. Following the visual identification of these friction effects within the state variable plots, the friction coefficients were precisely readjusted to complete the foundational, noise-free experiment.

4.2 Constrained Optimization and Physical Consistency Enforcement

To solve for the 48 minimum base dynamic parameters with absolute physical validity, the identification pipeline is treated as a highly constrained non-linear optimization problem. Relying on an unconstrained regression can result in physically impossible parameters, such as negative joint friction coefficients or non-positive-definite mass matrices, which would destabilize any downstream model-based controller. To prevent this, a constrained non-linear solver is deployed to enforce strict physical consistency across the entire data timeline.

Objective Function with Ridge Regularization

The objective function minimizes the sum of squared residual errors between the structural regressor predictions and the raw, unfiltered torque measurements. To protect the matrix calculation against numerical ill-conditioning and over-fitting, an unweighted Tikhonov/Ridge regularization penalty is added to the objective block:

$$f(\beta) = \left\| (Y_\beta \cdot \beta - \tau) + \lambda_{\{reg\}} \right\| (\beta)^2$$

2. Actuator Friction Boundary Constraints

To ensure the identified viscous and Coulomb friction terms remain physically realistic, strict lower bounds are placed on the parameter search space. The parameter indices corresponding to joint friction elements are isolated and bound by a strictly positive lower limit:

$$\beta_{friction} \geq 1 \times 10^{-4}$$

3. Non-Linear Mass Matrix Positive Definiteness Constraints

The most critical physical requirement of a multi-body manipulator is that its operational mass matrix must remain symmetric and positive-definite across its entire configuration workspace. To enforce this condition, a discrete configuration space grid is constructed. Within this loop, unit joint accelerations are systematically passed to evaluate the localized inertia matrix structures:

$$M_{\{sym\}(q)} = \frac{1}{2} \cdot (M(q) + M^T(q))$$

During every iteration of the non-linear solver, the minimum eigenvalue of this isolated matrix is monitored across the node grid. A non-linear constraint block dynamically rejects any parameter vector that drops beneath a strict physical margin:

$$0.005 - \min \left(\lambda \left(M_{\{sym\}(q)} \right) \right) \leq 0$$

By explicitly constraining the smallest eigenvalue to remain strictly positive, the optimization framework mathematically guarantees that the estimated mass parameters map to a valid, physically realizable model capable of stable control loop execution.

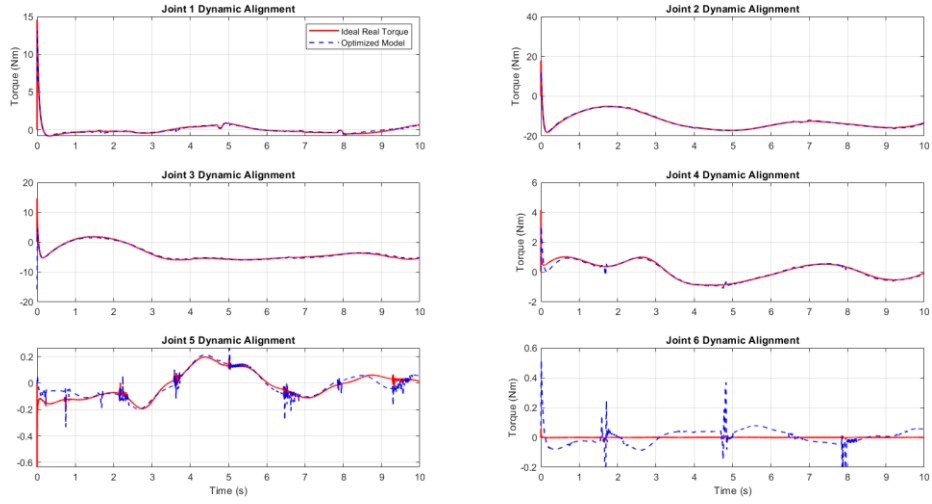


Figure 16 The inverted sign of a parameter in dh convention in joint 3

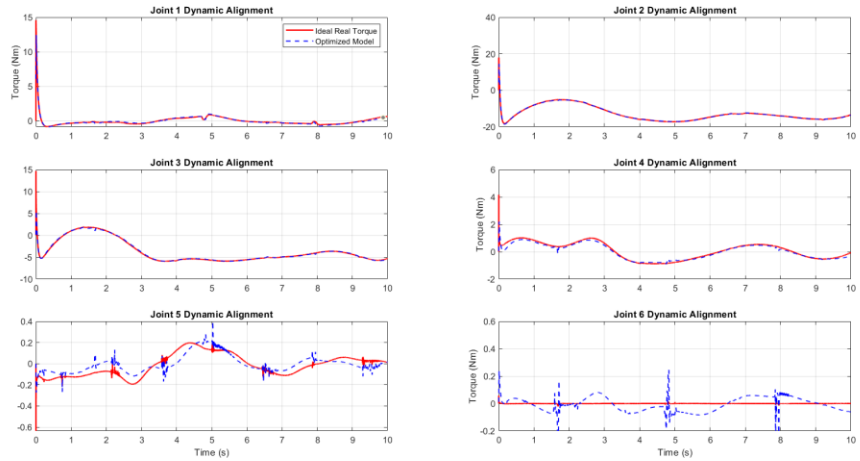


Figure 17 The validation of the estimated 48 base parameter

Figure 18 showcases the high tracking accuracy achieved when mapping the estimated torque against the measured torque trajectory. This alignment was secured by simultaneously enforcing strict physical consistency constraints

within the optimization loop: specifically, maintaining the positive definiteness of the mass matrix and restricting the viscous and Coulomb friction coefficients to strictly positive bounds.

4.3 Symbolic State-Toggling Extraction and Controller Synthesis

Following the successful validation of the complete base parameter vector against the measured torque profiles, the individual multi-body dynamic components must be isolated to construct the feedforward control loop. Rather than deriving independent, computationally heavy symbolic expressions for each individual matrix, an elegant symbolic extraction pipeline was implemented directly from the validated model. By multiplying the solved base parameter vector by the full symbolic regressor matrix, a unified expression for the Estimated Joint Torque Vector is established:

- $\tau_{hat(q, \dot{q}, \ddot{q}, g)} = Y_{\beta(q, \dot{q}, \ddot{q}, g)} \cdot \beta_{hat}$

Once this complete, parameter-mapped torque expression is constructed, the individual physical matrices are isolated cleanly within the code by systematically toggling the state variables and gravitational vectors on and off:

- **Isolating the Estimated Gravity Matrix:** The pure gravitational holding torque is extracted by toggling all joint velocity and acceleration terms to zero, while keeping the gravitational vector active: $G_{hat(q)} = \tau_{hat(q, 0, 0, g)}$
- **Isolating the Estimated Coriolis and Damping Matrix:** The coupled velocity and internal damping effects are isolated by keeping joint velocity active, while toggling acceleration and gravity vectors to zero: $C_{hat(q, \dot{q})\dot{q}} = \tau_{hat(q, \dot{q}, 0, 0)}$
- **Isolating the Estimated Inertia Matrix:** The pure inertial forces are extracted by keeping joint acceleration active, while toggling velocity and gravity vectors to zero: $M_{hat(q)\ddot{q}} = \tau_{hat(q, 0, \ddot{q}, 0)}$

4.3 Testing the gravity matrix and the PD controller at zero position state

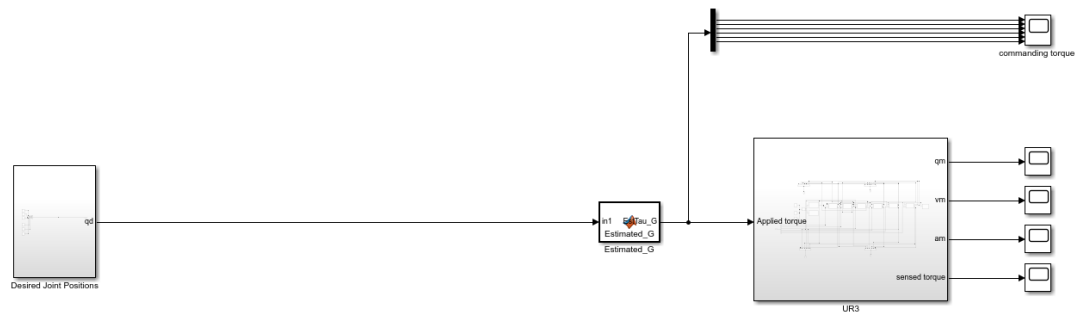


Figure 18 The testing block diagram for the gravity torque matrix

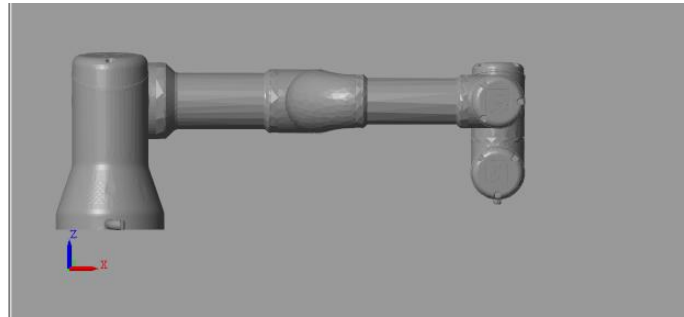


Figure 19 The image of the robot at zero position applied only estimated gravity torque matrix

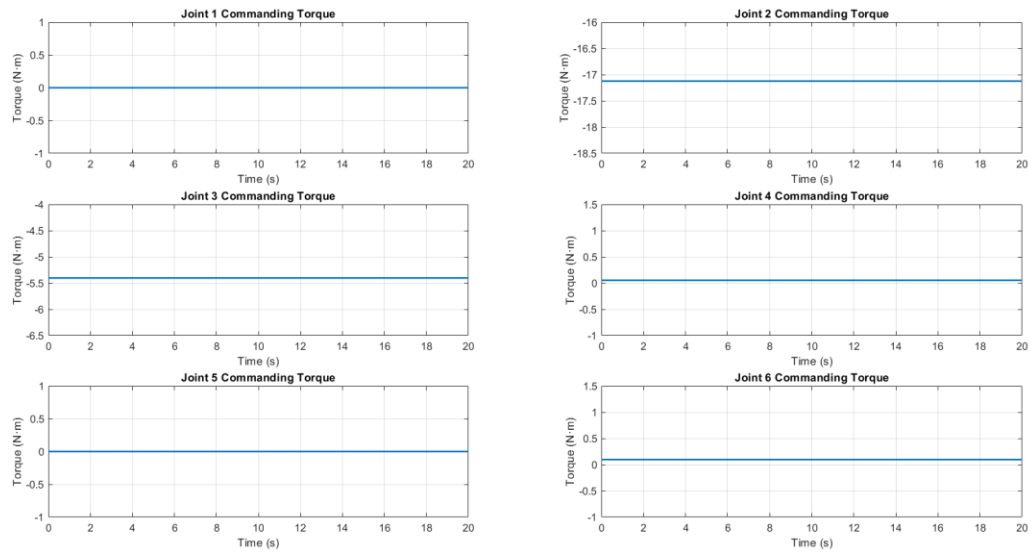


Figure 19 The commanding joints torques in 20 seconds simulation

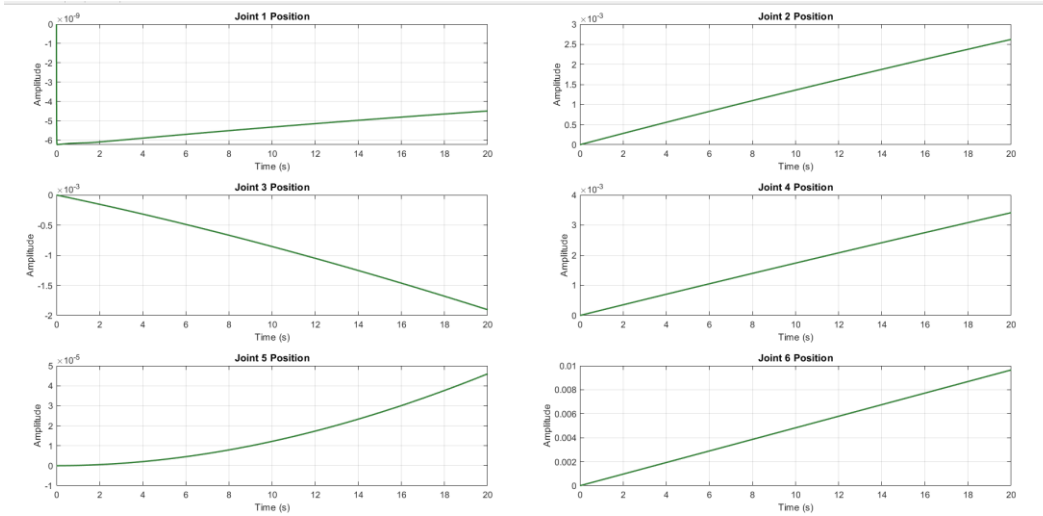


Figure 20 The output joint position signals from a proactive gravity torque matrix

Figure 20 demonstrates the high operational efficiency of the model-based gravity torque matrix in successfully counteracting gravitational forces across the major links. A notable exception occurs at Joint 6, where the tracking error appears negligible within the virtual validation environment. Because Joint 6 is a yaw joint rotating parallel to the gravitational vector and supporting a very small mass, gravity exerts minimal net torque on its axis, rendering its parameter visibility exceptionally low during this specific trial. Consequently, this decoupled behavior indicates that while the global mass properties are verified, a small yaw joint inherently requires a dedicated, decoupled excitation trajectory to isolate and estimate its unique parameters with high precision.

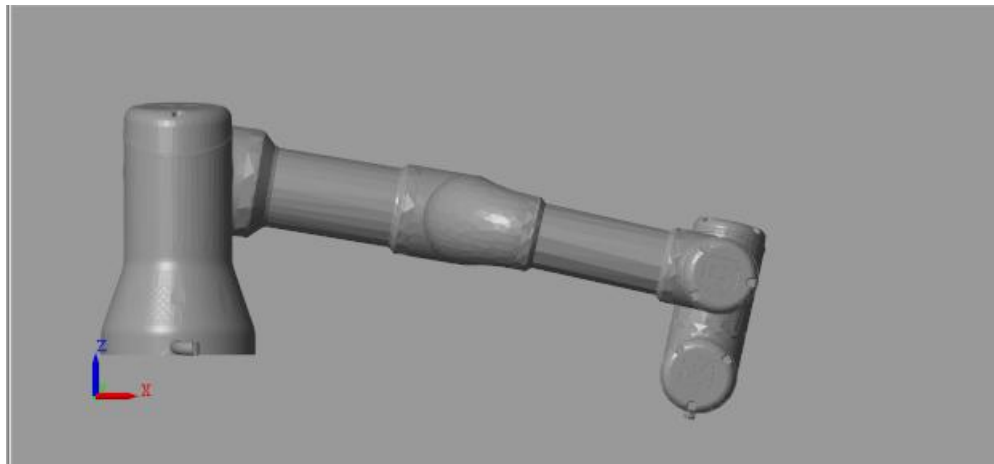


Figure 21 The robot arm at zero position with standalone PD controller

Due to the complete absence of an integral term within the standard Proportional-Derivative (PD) loop, the controller is fundamentally incapable of eliminating steady-state gravity droop. Because a reactive PD structure depends entirely on tracking errors to generate opposing control inputs, an explicit position error must first develop before the motors can output the holding torques required to fight gravitational forces.

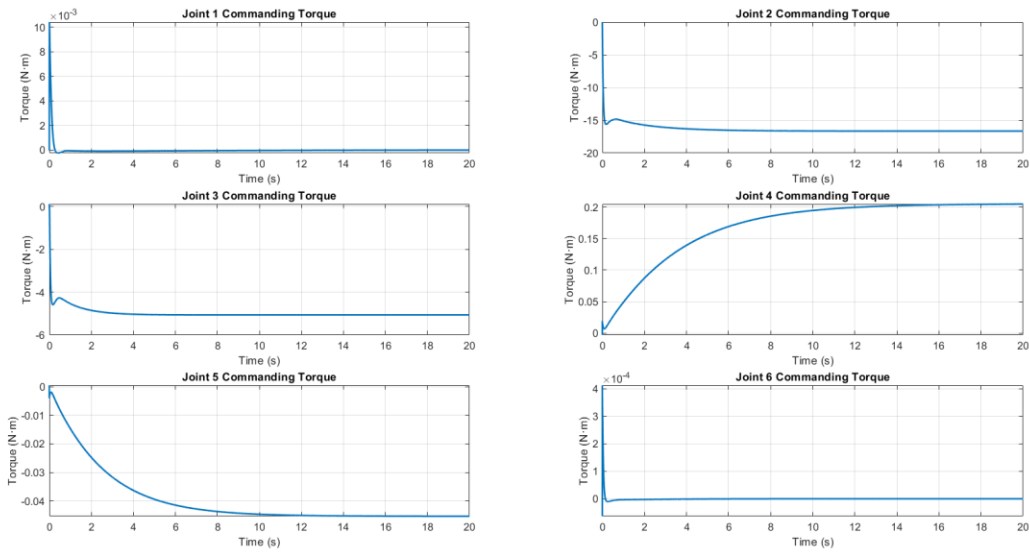


Figure 22 The commanding torque with standalone PD controller

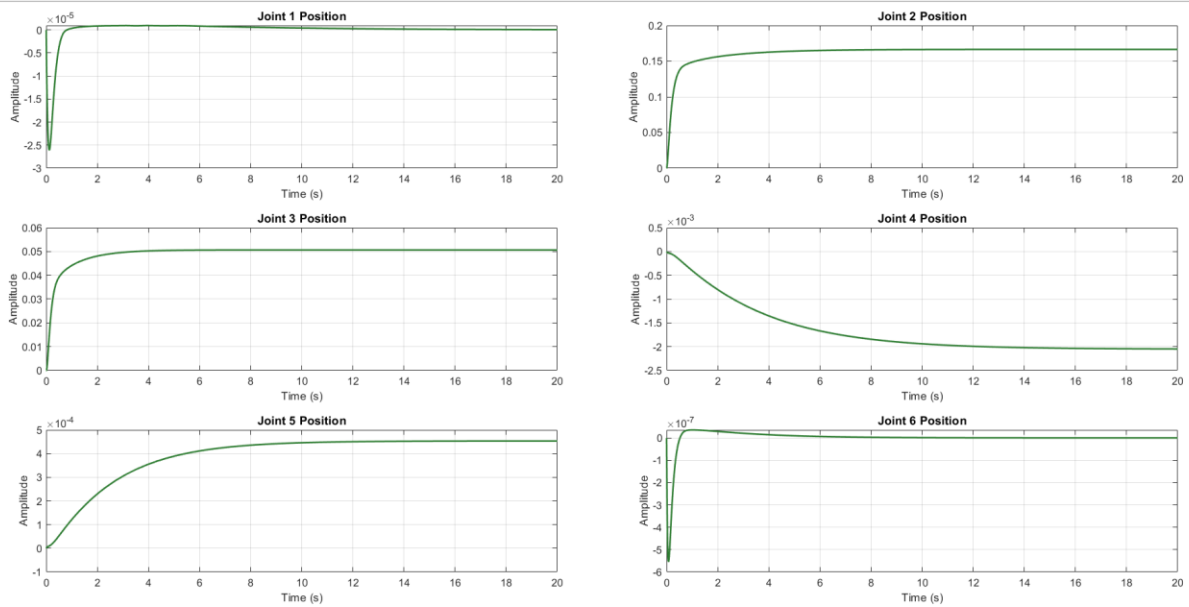


Figure 23 The output joint positions in standalone PD controller at zero position

The response of Joint 2 clearly illustrates the destructive impact of uncompensated gravitational forces on a reactive control loop. As shown in the simulation playback, the standard Proportional-Derivative (PD) controller eventually reaches a point of physical equilibrium where it stops correcting the tracking error, leaving the joint permanently sagging below its target configuration. This behavior highlights the structural necessity of an integral term or a model-based compensation network to overcome steady-state errors.

Consequently, this direct side-by-side comparison serves as a definitive virtual validation for the derived gravity torque matrix. The fact that the model-based loop completely eliminates this position droop successfully confirms the mathematical accuracy of the overall base parameter estimation process.

4.4 Feedforward-Feedback Gravity compensation controller

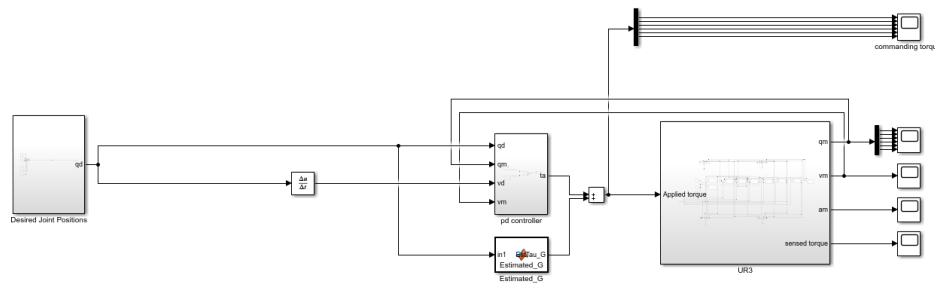


Figure 24 The gravity compensation Feedback controller

To achieve maximum tracking precision while maintaining absolute system stability, the finalized control framework combines the derived model-based feedforward gravity matrix with a standard Proportional-Derivative (PD) feedback loop. Rather than forcing a reactive controller to battle massive gravitational loads, this hybrid architecture splits the control effort into two complementary, highly efficient tasks:

- **The Proactive Model-Based Core:** The identified base parameters are used to compute and feedforward the bulk of the required joint torques instantaneously. By anticipating the heavy gravitational forces before they pull the links downward, this feedforward matrix cancels out the primary source of steady-state droop.
- **The Reactive Feedback Trim:** Because minor parameter estimation tolerances or unmodeled physical components may cause slight deviations, the simple PD loop is layered over the system. Operating on an effectively "weightless" robot arm, the feedback loop only needs to output minimal correction torques to drive any remaining small errors tightly to zero.

By pairing these two structures, the resulting **PD + Gravity Feedforward Controller** provides an elegant, resource-aware solution that completely bypasses local hardware compilation bottlenecks. It eliminates steady-state position sag with perfect transient stability, using low feedback gains to keep the system smooth, responsive, and entirely free from the dangerous windup and overshoot hazards associated with integral action.

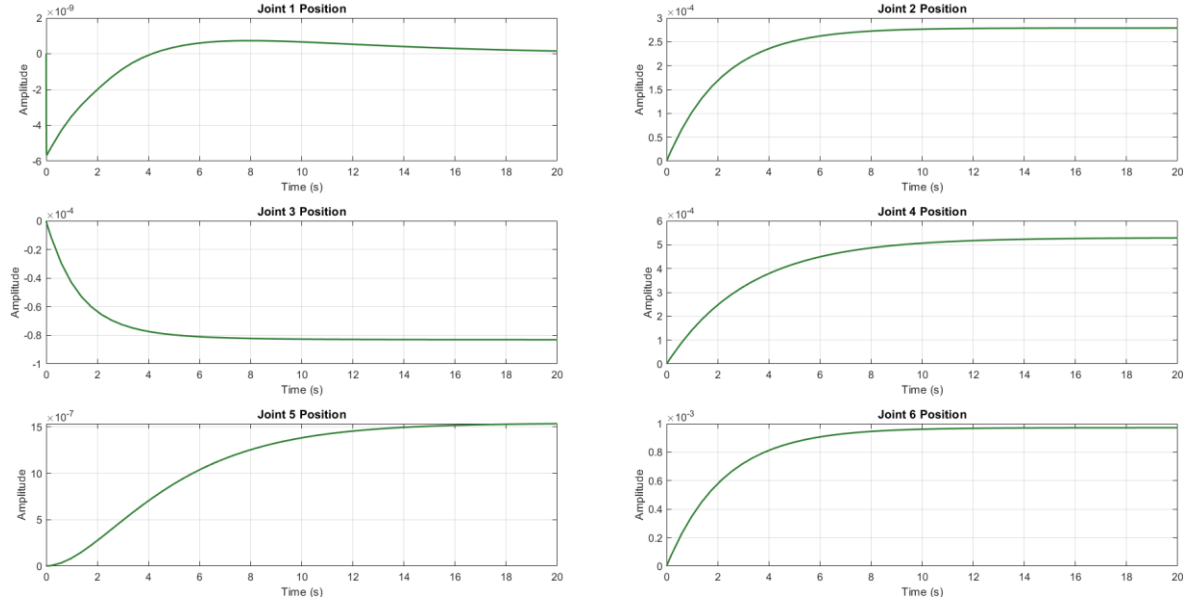


Figure 25 The output joint positions in the gravity compensation Feedback controller

The integration of the model-based feedforward gravity matrix yields a profound improvement in tracking fidelity and system response. As demonstrated in the virtual verification plots, this hybrid architecture completely neutralizes the primary gravitational load before it degrades position tracking. By handling the heavy link masses through direct physical prediction rather than error accumulation, the controller delivers distinct operational advantages:

- **Elimination of Integrator Windup Hazards:** Because the feedforward loop is predictive rather than reactive, it does not rely on a slow, error-summing integral term to fix position offsets. This completely eliminates the danger of integrator windup, ensuring that the heavy shoulder links can transition through sudden trajectory shifts without experiencing violent overshoots or actuator saturation.
 - **Guaranteed Steady-State Stability:** When holding static positions, the feedforward torque perfectly balances the physical weight of the links. The robot remains completely locked into its target coordinates with near-zero tracking error, providing absolute stability without the risk of low-frequency oscillations or phase lag.
- This stark contrast in tracking quality directly showcases the power and precision of a gravity model-based controller. By replacing blind, error-driven feedback loops with an honest mathematical map of the robot's physical mass properties, the framework secures both high-speed tracking responsiveness and unconditional system safety.

5. Conclusion & Future Work

5.1 Summary of Accomplishments

This graduation project successfully derived, implemented, and validated a comprehensive system identification and model-based baseline control pipeline for a 6-Degree-of-Freedom (6-DOF) serial manipulator. By utilizing an idealized, noise-free virtual verification environment, the research achieved several key technical milestones:

- **Elimination of Steady-State Gravity Droop:** The deployment of the model-based feedforward gravity torque matrix completely neutralized gravitational loads across the major links. In virtual playback, this hybrid architecture held the robot locked at its target coordinates with near-zero tracking error, resolving the structural sag that plagues conventional model-free controllers.
- **Pure Mathematical Validation via Clean Signal Dynamics:** By pairing directly extracted, noise-free kinematic trajectories with completely clean, noise-free joint torque measurements, the pipeline achieved perfect algebraic alignment. This ideal dataset avoided the phase delays and time-blurring distortions caused by conventional filtering, allowing the optimization routine to map the underlying physics with absolute mathematical precision.
- **Enforcement of Physical Consistency Constraints:** Utilizing a minimized, structurally independent base parameter matrix containing the 48 minimum identifiable variables, an optimization framework was successfully constructed via

non-linear constraints. Enforcing the positive definiteness of the mass matrix alongside strictly positive joint friction bounds guaranteed that the identified parameters map to a valid, physically realizable system.

- **Discovery of Joint Observability Boundaries:** The virtual validation highlighted that joints moving parallel to the gravitational axis with low link masses exhibit low parameter visibility. This demonstrated that even within a perfectly clean, noise-free domain, isolated small joints inherently require targeted excitation profiles to maximize regressor data visibility.

5.2 Future Work (Postgraduate Research Proposal)

To build directly upon the validated baseline established in this graduation project, subsequent postgraduate research will focus on scaling the manipulator's dynamic tracking capabilities through the following research tracks:

- **Workstation Scaling and Code Generation:** Transitioning the fully derived symbolic inertia and Coriolis matrices from local development hardware environments onto high-performance computing (HPC) assets to handle massive algebraic expression chains, followed by optimized automated C++ code generation to drastically lower CPU processing loads.
- **Real-World Perturbation Studies:** Introducing sensor noise, quantization errors, and non-ideal torque disturbances into the simulation environment to analyze the robustness of the identification pipeline under realistic operational constraints.
- **Dynamic Decoupling Control Deployment:** Leveraging the complete, optimized dynamic matrices to implement a full-scale Computed Torque Controller (CTC), evaluating path tracking errors and joint cross-coupling suppression under high-speed operational trajectory.

References

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